
Robust Low-Light Image Enhancement via Generative Priors: A CSCE753 Project Proposal

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Abstract

Low-light image enhancement is a critical task in computer vision, with applications spanning surveillance, autonomous vehicles, medical imaging, and remote sensing. While deep learning-based approaches have significantly advanced the field, existing generative models often struggle with domain adaptation to real-world conditions, particularly in extreme darkness and non-uniform illumination scenarios.

This project aims to develop a novel low-light image enhancement method tailored for the NTIRE 2025 Challenge on Low-Light Enhancement, focusing on improving brightness, clarity, and perceptual quality while minimizing artifacts. This work will systematically explore and benchmark existing generative foundation models (e.g., GANs, diffusion models, transformers) to identify performance gaps and limitations. The developed solution will prioritize flexibility, ensuring future adaptability for different applications where enhancing low-light plant imagery is crucial for downstream analysis.

1 Introduction

Image manipulation is a key computer vision task, aiming at the restoration of degraded image content, the filling in of missing information, or the needed transformation and/or manipulation to achieve a desired target (with respect to perceptual quality, contents, or performance of apps working on such images). Recent years have witnessed an increased interest from the vision and graphics communities in these fundamental topics of research. Not only has there been a constantly growing flow of related papers, but also substantial progress has been achieved.

Each step forward eases the use of images by people or computers for the fulfillment of further tasks, as image manipulation serves as an important frontend. Not surprisingly then, there is an ever growing range of applications in fields such as surveillance, the automotive industry, electronics, remote sensing, or medical image analysis etc. The emergence and ubiquitous use of mobile and wearable devices offer another fertile ground for additional applications and faster methods. [1].¹

In this context, generative foundation models —ranging from GANs and transformer-based models to the more recent diffusion-based approaches such as those seen in Flux-1.dev or SDXL— offer a promising solution for overcoming the challenges posed by real-world low-light conditions. These models are pre-trained on vast and diverse datasets, enabling them to learn rich priors about natural image structures and textures. By adapting these large-scale models to the low-light enhancement task, our project seeks to harness their inherent strengths to improve both the perceptual quality of enhanced images and the reliability of subsequent high-level vision applications. The goal is to

¹See also the NTIRE Low Light Image Enhancement Challenge at CVPR 2025: <https://codalab.lisn.upsaclay.fr/competitions/21636>.

bridge the gap between state-of-the-art generative techniques and practical, real-world scenarios characterized by extreme darkness and uneven illumination, thereby paving the way for more robust machine perception in fields such as surveillance, autonomous navigation, and remote sensing.

Despite the significant progress in low-light image enhancement using generative models, several critical challenges remain that hinder their deployment in real-world scenarios. First, *domain adaptation* is a persistent issue, as models pre-trained on synthetic or controlled datasets often struggle to generalize to real-world conditions characterized by diverse lighting patterns and sensor noise. Second, the *efficiency constraints* of diffusion-based approaches pose a challenge for real-time applications, especially on resource-constrained hardware such as those in autonomous vehicles or robotics. Additionally, maintaining *structural integrity* is crucial; it is essential to avoid artifacts and preserve semantic details to ensure safe and reliable performance in high-stakes applications like autonomous driving. Finally, while *multi-modal integration* (incorporating modalities such as depth information or segmentation maps) can significantly boost performance, it also complicates network design and training.

2 Related Work

Recent years have witnessed a surge of interest in low-light image enhancement, driven by its critical role in a variety of applications ranging from autonomous driving to remote sensing and medical imaging. Generative models have become a cornerstone in this area, with several approaches proposed to address the unique challenges posed by extreme darkness and non-uniform illumination.

Diffusion-based frameworks have emerged as a powerful paradigm for unsupervised low-light enhancement. For instance, [2] introduces *Light the Night*, a multi-condition diffusion framework specifically designed for unpaired low-light enhancement in autonomous driving. Similarly, [3] proposes *LightenDiffusion*, which leverages latent-Retinex diffusion models to effectively enhance low-light images without the need for paired supervision. These methods exploit the iterative refinement capability of diffusion models to progressively improve image brightness and clarity.

On the other hand, GAN-based methods continue to play an important role. [4] presents *EnlightenGAN*, a deep learning approach that performs low-light enhancement without paired supervision by employing adversarial training to recover realistic lighting and color. In parallel, super-resolution methods have also been adapted for low-light scenarios. For example, *RealESRGAN* [5] was originally developed for real-world image super-resolution and has been extended to improve visual quality and denoising in low-light conditions.

Benchmark datasets are crucial for evaluating these models. General low-light repositories such as LOLv1[6], LOLv2[7], and SID[8] provide extensive testbeds for visual enhancement quality and denoising performance. For application-specific evaluation, real-world driving datasets like nuScenes[9], BDD100K[10], and Dark Zurich[11] offer diverse nighttime scenarios to assess both visual clarity and the impact on downstream perception tasks such as object detection. In addition, synthetic simulations generated by platforms such as CARLA[12] or GTA V [13] enable controlled evaluations and facilitate domain adaptation studies.

These works underscore the potential of generative foundation models for low-light enhancement while highlighting key challenges: robust domain adaptation, computational efficiency, preservation of structural integrity, and effective multi-modal integration. Our project builds on these insights by focusing on robust domain adaptation—an achievable goal for this project—to ensure that the enhanced images generalize well to real-world low-light conditions, thereby improving both human perceptual quality and downstream task performance.

3 Evaluation

To assess the performance of our low-light image enhancement method, we employ both quantitative and efficiency metrics. In particular, we focus on two widely-used image quality metrics—Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM)—to evaluate the visual fidelity of the enhanced images. In addition, we study the computational complexity of our approach, measuring aspects such as inference time, memory usage, and overall resource demands, which are critical for deployment in real-time, resource-constrained applications.

3.1 Quantitative Metrics

3.1.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the ratio between the maximum possible power of a signal (i.e., the image) and the power of corrupting noise. It is expressed in decibels (dB) and is computed as:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\text{MSE}} \right), \quad (1)$$

where L is the maximum possible pixel value (e.g., 255 for an 8-bit image) and MSE is the mean squared error between the enhanced image and the corresponding ground truth image. Higher PSNR values indicate better reconstruction quality.

3.1.2 Structural Similarity Index (SSIM)

SSIM is a perceptual metric that assesses the similarity between two images based on luminance, contrast, and structural information. It is designed to be more aligned with human visual perception than traditional metrics like PSNR. The SSIM between images x and y is defined as:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (2)$$

where μ_x and μ_y are the means, σ_x^2 and σ_y^2 are the variances, and σ_{xy} is the covariance of x and y . The constants C_1 and C_2 stabilize the division when the denominators are small. SSIM values range from 0 to 1, with higher values indicating greater similarity to the ground truth.

3.2 Computational Complexity

In parallel with visual quality evaluation, we analyze the computational complexity of the proposed method. This evaluation includes measuring the inference time and resource utilization (e.g., memory consumption and processing power) during both training and testing. Assessing computational demands is crucial for ensuring that our approach can be feasibly deployed on resource-constrained platforms, such as those in autonomous vehicles or robotics, where real-time performance is essential.

Overall, the combination of PSNR and SSIM provides a rigorous quantitative assessment of image enhancement quality, while the computational complexity analysis ensures that the developed method is practical for real-world applications.

4 High-Level Project Plan

The project is structured into three major phases to meet the NTIRE 2025 Challenge deadlines while ensuring flexibility for future extensions (e.g., application to plant phenotyping). The following high-level plan outlines the primary milestones:

4.1 Phase 1: Development and Baseline Establishment (Feb 21 – Mar 7)

- **Data and Environment Setup:**
 - Acquire the NTIRE low-light training dataset and complementary public datasets (e.g., LOL, LOL-v2).
 - Configure the development environment with necessary libraries and GPU resources.
- **Model Survey and Baseline Implementation:**
 - Review recent literature to select 2–3 promising generative foundation models (e.g., diffusion-based and GAN-based approaches) for low-light enhancement.
 - Develop baseline implementations and run initial experiments to establish performance benchmarks on cases with extreme darkness and non-uniform illumination.

4.2 Phase 2: Testing and Refinement (Mar 7 – Mar 21)

- **Testing and Evaluation:**
 - Evaluate baseline models on the NTIRE test set using quantitative metrics (PSNR, SSIM, LPIPS, NIQE) and qualitative assessments.
- **Algorithm Refinement:**
 - Identify performance gaps and implement targeted enhancements (e.g., refining brightness/contrast modules, optimizing loss functions to reduce artifacts).
 - Conduct comparative analysis and ablation studies to validate the improvements.
- **Optional Domain Extension:**
 - If time permits, adapt the enhanced model for a small plant image dataset to simulate low-light plant phenotyping, showcasing potential improvements in downstream segmentation tasks.

4.3 Phase 3: Finalization and Reporting (Mar 21 – Late April)

- **Documentation and Analysis:**
 - Consolidate experimental results and insights into a comprehensive final report.
 - Prepare visual comparisons (e.g., graphs, qualitative results) to clearly illustrate the model's performance improvements.
- **Presentation and Future Work:**
 - Develop a final presentation summarizing the objectives, methodology, results, and potential future extensions (with a focus on integrating plant segmentation for phenotyping applications).

5 Task Assignment

Currently, I am undertaking this project individually, handling all aspects including literature review, model implementation, experimentation, and analysis. However, I remain open to collaborating with other classmates who are interested in related topics. If opportunities for team merging arise, responsibilities may be redistributed to include additional contributions such as data processing, model tuning, evaluation, and documentation.

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